BMI Calculator Project

Functions

Function 1 — get_details()

Purpose

Collects the user's basic details and measurements from the keyboard.

Inputs

Name, age, gender, height in centimetres and weight in kilograms.

Return Value

Returns five values: name, age, gender, height and weight.

Key Python Concepts

input(), int(), float(), variables and return statements.

Function 2 — calculate_bmi(height, weight)

Purpose

Converts height from centimetres to metres and calculates BMI.

Formula

BMI = weight / (height × height)

Return Value

Returns the rounded BMI and the converted height in metres.

Function 3 — show_result(name, age, gender, height, weight, bmi)

Purpose

Displays the calculated result and applies the program's conditional logic for BMI category, suggestion and age group.

BMI Conditions

The function checks the BMI using an if/elif/else structure. Values below 18.5 are assigned to Underweight; values below 25 to Normal Weight; values below 30 to Overweight; and all remaining values to Obese.

Age Conditions

The function checks age separately. An age below 13 is labelled Child, below 20 is labelled Teenager, below 60 is labelled Adult, and 60 or above is labelled Senior.

Main Program Loop

Outside the functions, the program uses a while True loop to repeat the calculation. The user can choose “yes” to calculate again or “no” to end the program.

Validation

Before BMI is calculated, the program checks whether height and weight are greater than zero. Invalid non-positive values cause the program to return to the details section.